

Steven Ford's Reflection

I've experienced what the term vibe coder means through this project. It's used to describe someone who relies on AI to write their code which is what I had to do for each step of the way. It was a lot of plug and play, where I'd ask for each step of code and copy paste it into the notebook and then see if it works. My group tasked me to run smote to balance the dataset, but when I tried to run it, it prompted me to fix nominals to numerals and fill in missing data meaning that the dataset had to be treated first before it could be balanced. Gemini had me split the set as well into training and testing. It took me a few hours to get everything done with the only frustrating part was trying to push the files to the repo for some reason it wasn't working properly.

Khin Nyein Chan Thu

Working on this midterm project helped me understand how a raw dataset needs to be prepared before it can be used for machine learning. I learned how to inspect the dataset for missing values, duplicates, data types, and class distribution. One of the most important things I learned was that we need to split the dataset into 70% training and 30% testing before doing EDA. We performed EDA only on the training data so that the testing data would not affect our decisions. I also learned how to use Matplotlib to visualize the data and identify problems. In our stroke dataset, I noticed that there were many more non-stroke cases than stroke cases, which showed that the dataset was highly imbalanced.

I also learned different preprocessing techniques such as filling missing BMI values, scaling and normalizing numerical data, one-hot encoding categorical data, and creating new features through feature engineering. Another important part was learning how SMOTE can help balance the training dataset. Before SMOTE, there were 3,403 non-stroke cases and only 174 stroke cases in the training data, but after SMOTE both classes had 3,403 records. I learned that SMOTE should only be applied to the training data and not the testing data. Overall, this project helped me understand that data cleaning and preprocessing are very important steps before building a machine learning model, and I now feel more comfortable using Python and Jupyter Notebook to prepare data for future ML projects.

Cristian Delgado's Reflection

Upon searching for datasets compatible within the scope of our midterm requirements, our team discovered that data gets collected and stored in a wide variety of formats. Often, datasets relating to the same topic will have been cataloged in different manners with no standardization. As such, a challenge we faced was finding a dataset that had a balance between being simple enough to work with for our first machine learning project while still containing enough features capable of providing useful insight. Additionally, as part of our project required being able to show effective cleaning of data, we had to find a dataset that hasn't been processed yet.

We eventually settled on a dataset that focused on stroke indicators, with the expectation that we will be able to assign a form of risk score based on features presented by the patient. The cleaning itself was relatively straightforward. Though I was not in charge of majority of the coding, I did create my own version of the cleaned csv in case it was needed. My backup was created with Google Gemini providing most of the code through prompts and was looked over after each code cell to make sure that the data was modified only in the manner requested and was repeatable. While my version was not needed in the end, it proved useful in understanding the process that goes into cleaning a dataset to make it useable in a machine learning model.

One question I have is how a more feature-rich dataset would have impacted the cleaning process. Based on the lessons in class, the cleaning process would have required certain features to be modified in a particular manner in order for the algorithms to train on it, greatly expanding the parameters and the table. For learning, this might prove harder to work with and thus more undesirable, but the tradeoff would be more features for the model to learn on. Additionally, how much more accurate (or possibly inaccurate) would a model with more parameters be compared to a simpler one. If a model was trained on one dataset and a different one trained on a different

set with less features but an equal amount of entries, and then employed to review patient data, would the risk scores be comparable or would the extra features provide significantly greater insight in this application.

Antonio Garcia Reflection

For this midterm, our group worked with a stroke prediction dataset from Kaggle. My main responsibility was handling the categorical encoding for two important features: `smoking_status` and `Residence_type`. Before I started, I didn't realize how much encoding affects the entire machine learning pipeline. Once I began working with the dataset, I saw that these columns had multiple categories, including values like "formerly smoked," "never smoked," "smokes," and "Unknown," as well as "Urban" and "Rural" for residence type. Because machine learning models can't work with text categories directly, converting them into numerical values was essential.

I learned how to choose the right encoding method depending on the type of category. `Residence_type` only had two values, so label encoding made sense. `Smoking_status` had four categories, so I used one hot encoding to avoid forcing an order or ranking that doesn't exist. While doing this, I also had to pay attention to how encoding interacts with the rest of the pipeline. For example, SMOTE requires numeric features, so encoding had to be done before balancing the dataset. I also had to make sure the encoded columns wouldn't cause issues later during scaling or model training.

Another part of my contribution was helping confirm that our dataset was a good

choice for the project. The stroke dataset has mixed data types, missing values, and a strong class imbalance, which fits the midterm requirements perfectly.

Understanding why the dataset works helped me see how preprocessing, encoding, and balancing all connect. Even though my task focused on encoding, I gained a better understanding of how each team member's part fits together to create a complete machine learning workflow.

Overall, this assignment helped me understand the importance of preprocessing and how small steps like encoding can impact the final model. I also learned how to collaborate with my group by taking ownership of a specific part while still staying aware of the bigger picture. Encoding may seem simple at first, but it plays a major role in preparing the dataset for SMOTE and for training a model that can actually learn meaningful patterns.